

Data Mining in the Network Era
G0303
Welcome

Longbiao CHEN

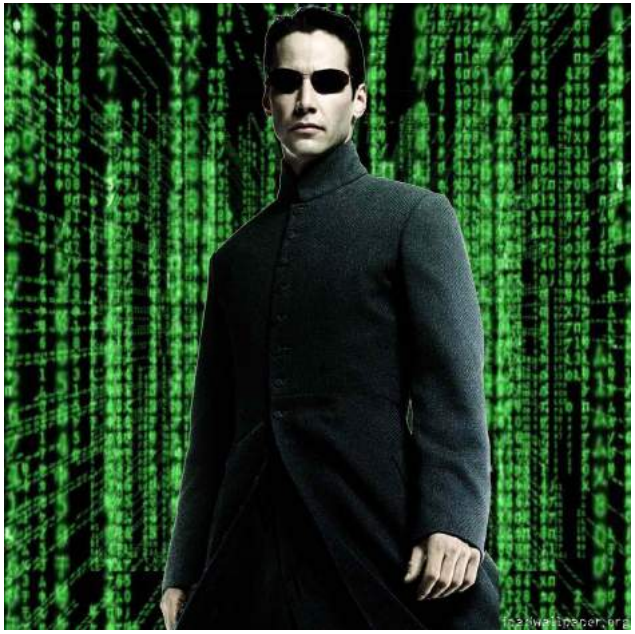
Xiamen University Malaysia

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About Me

- **Longbiao CHEN (陈 龙彪)**
- **Assistant Professor**, Xiamen University, China
- **Ph.D.:** University of Paris VI, Paris, France
- **M.S. and B.S.:** Zhejiang University, China
- **Research:** Data Mining, Network Analytics, Intelligent Transportation Systems (ITS)

About the Course





People think data mining is like… while actually it is like…

Image courtesy: The Matrix (1999)

About the Course

- **G0303**: Data Mining in the Network Era
- **5 weeks:** lecture and Practice
  - **Lecture**: twice/week, TUE/THU, room 104
  - **Practice**: once/week, FRI, room 106
- **1 week:** examination (no mid-term examination 😄)
- **Website**: http://longbiaochen.com/courses/g0303/

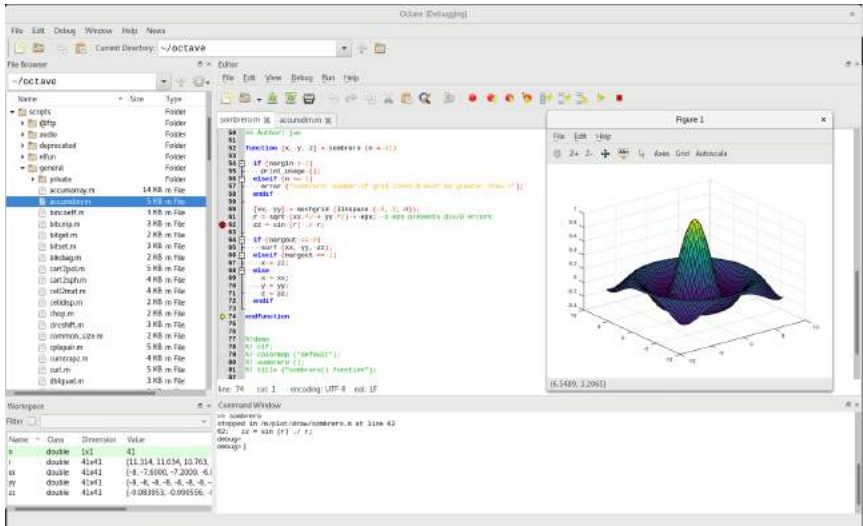


Lecture (Room 104)

- **Length**: 10 lectures (2*5), 3 hours per lecture
- **Rules**: casual, walk-in/out, ask questions, ~~*sleep*~~
- **Syllabus**
  - Week 1: Introduction and Data Exploration
  - Week 2: Classification Techniques
  - Week 3: Association Analysis
  - Week 4: Clustering Techniques
  - Week 5: Anomaly Detection and Review
- **Assignment**: 3 times (week 2, week 3, week 4)

Practice

- **Length**: 5 weeks, 3 hours per week
- **Objective**
  - learning to use Octave/Matlab for data mining
- **Syllabus**
  - Week 1: Installing Octave
  - Week 2: Octave Basics
  - Week 3: Flows and Functions
  - Week 4: Matrices
  - Week 5: Graphs



Assignment

- Three times: week 2, 3, 4
- **Download:** http://longbiaochen.com/courses/g0303/
- **Hand in**:
  - With formal cover page
  - Signed
  - One-side printed



Consultation

- **Length**: 2 hours each time, twice per week
- **Time**: 20:00—22:00, TUE and THU
- **Venue**: Academic Office **108E**

Examination

- **Questions**: 6 questions, 2 hours
- **Closed book**: calculator allowed, no other materials
- **Scope**: based on assignments and lectures

Grading

- **Overall Marks** = 50% Assignment + 50% Exam
  - Assignment #1: 10 marks
  - Assignment #2: 20 marks
  - Assignment #3: 20 marks
  - Exam : 50 marks (scaled)
- **Attendance record**: signature in person, week 2–4
  - Required by MQA for attending the exam

References

- **Textbook**
  - Pang-Ning Tan, Michael Steinbach, Vipin Kumar, **Introduction to Data Mining**. Addison Wesley, 2006.
    - http://www-users.cs.umn.edu/~kumar/dmbook/index.php
- **Related**
  - Andrew Ng. **Machine Learning.** Coursera.org
    - https://www.coursera.org/learn/machine-learning
  - Jiawei Han. **Pattern Discovery in Data Mining**. Coursera.org
    - https://www.coursera.org/learn/data-patterns

Contact Me

- Longbiao CHEN (陈 龙彪)
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